

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ANNA UNIVERSITY, CHENNAI - 600025

BE-CSE VI SEM - N, P & Q Batches

Date: 02-02-2026

Max Marks: 50

Time: 1.30 mins

PART A (10*2=20 Marks)

1. What are the components of a compiler front end?
2. Compare compiler vs interpreter with respect to execution.
3. Consider the context-free grammar
$$S \rightarrow S S + | S S * | a$$

Show how the string $aa+a^*$ can be generated by this grammar.
4. Write a Regular Expression for a valid email ID (basic format).
5. What language is generated by the following grammar?
$$S \rightarrow a | S + S | S S | S * | (S)$$
6. What is panic mode error recovery in lexical analysis?
7. Remove left recursion in the following grammar

$$\begin{aligned} S &\rightarrow A B \\ A &\rightarrow A a | \epsilon \\ B &\rightarrow B b | c \end{aligned}$$

8. Define Lexeme and Pattern.
9. Differentiate Buffer pairs and Sentinels.
10. Consider the context-free grammar
$$E \rightarrow E + E | E * E | id$$

Perform handle pruning for the string $id + id + id$.

PART B (30 Marks)

ANSWER ALL QUESTIONS

11. a. Is grammar LL(1)? (10)
$$\begin{aligned} S &\rightarrow A a | b \\ A &\rightarrow A c | S d | \epsilon \end{aligned}$$

(OR)

b. Is grammar LL(1)?
$$\begin{aligned} S &\rightarrow i E t S | i E t S e S | a \\ E &\rightarrow b \end{aligned}$$
12. Show that the grammar is CLR(1) but not SLR: (15)
$$\begin{aligned} S &\rightarrow L = R | R \\ L &\rightarrow *R | id \\ R &\rightarrow L \end{aligned}$$
13. For the above grammar (Q.No. 12) construct LALR parsing table. (5)

*****ALL THE BEST*****

$$\begin{aligned} S &\rightarrow A a | b \\ A &\rightarrow S d A' | A' \\ A' &\rightarrow c A' | \epsilon \end{aligned}$$

Date: 26-03-2026

Max Marks: 50

Time: 1.30 mins

PART A (10*2=20 Marks)

PART B (30 Marks)

ANSWER ALL QUESTIONS

1. a. Attributes computed using only child nodes are called _____ attributes.
b. A parse tree where each node has computed attribute values is called _____
2. Write short circuit code for if (a==b || c==d || e==f) x = 1
3. Write semantic action for type conversions into expression evaluation.
4. Write the semantic rules for constructing syntax tree for simple expressions.
5. Write the type expressions for array, function, pointer and record.
6. Write SDT for repeat-until (do-while) loop structure
7. Translate the expression (a*-b)+e^2 to 3 address code in Triples
8. Compute liveness and next-use information for the following basic block.

a=b+c
d=a+b

9. Eliminate the local common sub expression in the following basic block using DAG

1) t1 = a + b
2) t2 = c - d
3) t3 = a + b
4) t4 = t3 * e
5) t5 = c - d
6) t6 = t5 + t4

10. What is the significance of next-use information in register allocation?

PART B (3*10=30 Marks)

Answer any 3 Questions

11. Write the SDT rules and draw the annotated parse tree with relevant SDT scheme for the following statement.

A = B[i, j] + D[k]

Integer values held at B[i,j]=45, D[i]=78, D[k]=24, Sizes of the arrays are: B =40x40, D =50, i=j=k=25, assume base addresses.

12. Store the ICG form of the high level code in quadruple where set is a function with arguments and p is a single dimension array

S = set (a, 9)

A = S + p[i]

13. Using Backpatching of Boolean Expression and control statements translation technique convert the following if condition to 3-address code. Assume start address at 100. Draw the parse tree.

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a = -c * d
if (((a > b) && !(c == d)) || (g))
    a = f * a
else
    a = b++

```

14. Construct the flow graph for the following three address code, identify the loops in the flow graph and identify local common sub expressions (if any) for each loop.

1) $i = 1$	17) $t_{13} = t_{12} - 88$
2) $j = 1$	18) $t_{14} = b[t_{13}]$
3) $t_1 = 10 * i$	19) $t_{15} = t_9 * t_{14}$
4) $t_2 = t_1 + j$	20) $t_{16} = 10 * i$
5) $t_3 = 8 * t_2$	21) $t_{17} = t_{16} + j$
6) $t_4 = t_3 - 88$	22) $t_{18} = 8 * t_{17}$
7) $c[t_4] = 0.0$	23) $t_{19} = t_{18} - 88$
8) $k = 1$	24) $t_{20} = c[t_{19}]$
9) $t_5 = 10 * i$	25) $t_{21} = t_{20} + t_{15}$
10) $t_6 = t_5 + k$	26) $c[t_{19}] = t_{21}$
11) $t_7 = 8 * t_6$	27) $k = k + 1$
12) $t_8 = t_7 - 88$	28) <u>if $k \leq 10$ goto (9)</u>
13) $t_9 = a[t_8]$	29) $j = j + 1$
14) $t_{10} = 10 * k$	30) <u>if $j \leq 10$ goto (3)</u>
15) $t_{11} = t_{10} + j$	31) $i = i + 1$
16) $t_{12} = 8 * t_{11}$	32) <u>if $i \leq 10$ goto (2)</u>